# KOLA GANA SAI

**Software Engineer · Python & AI (Fresher)**

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## SUMMARY

Final-year Computer Science student who builds Python applications with FastAPI and applies NLP and machine learning to Indian-language problems, from multilingual voice assistants to Hinglish scam detection. Also trained in SAP ABAP, and is looking for an internship or entry-level software engineering or AI role.

## EDUCATION

**Chalapathi Institute of Engineering and Technology, Guntur** **2023 – 2027**

*B.Tech, Computer Science & Engineering | CGPA: 7.16/10*

**Sri Chaitanya Junior College — Intermediate (Class XII)** **811/1000 | 2023**

**DAV Public School — SSC (Class X)** **599/600 | 2021**

**Relevant coursework:** Data Structures, Database Management Systems, Object-Oriented Programming, Operating Systems, Computer Networks

## TECHNICAL SKILLS

**Languages:** Python, ABAP, SQL

**Python:** OOP, Flask, FastAPI, REST APIs, Data Structures, File & Exception Handling, pytest

**AI / ML & NLP:** scikit-learn, Text Classification, TF-IDF, BM25 Retrieval, Retrieval-Augmented Generation (RAG) basics

**SAP ABAP:** ABAP Dictionary (Domains, Data Elements, Tables), Internal Tables, Work Areas, Open SQL, Selection Screens, ALV Reports, Subroutines, Function Modules, SNRO Number Ranges

**SAP Data & Output:** CDS Views, OData, Smart Forms

**Tools:** SAP GUI, Eclipse ABAP Development Tools (ADT), VS Code, Git, GitHub

## PROJECTS

**Scheme Sahayak** — Multilingual Voice Assistant for Government Schemes GitHub

*Python, FastAPI, NLP, BM25 / RAG, JavaScript, Web Speech API*

- Built a voice assistant that answers welfare-scheme questions in Hindi, Marathi, Tamil and English, with speech input and spoken replies, served through a FastAPI back end.
- Combined BM25 retrieval over a 10-scheme multilingual knowledge base with a rule-based eligibility engine; an optional LLM only rephrases verified facts, so eligibility decisions stay explainable.
- Wrote rule-based extraction of age, occupation, land, BPL status and income from free text; correct language detection and a relevant scheme in the top 3 on all 24 queries of an internal test set, with 26 automated tests.

**UPI Scam Shield** — Offline Scam SMS Detector for Hinglish Messages GitHub

*Python, scikit-learn, FastAPI, Kotlin, Android*

- Trained a TF-IDF (word + character n-gram) and logistic-regression classifier with 20 fraud-rule features for English, Hinglish and Hindi SMS, reaching 0.94 F1 on message templates held out from training.
- Generated a 6,540-message synthetic dataset covering 11 scam types (UPI collect, fake KYC, digital arrest, job scams) with hard negatives such as genuine bank alerts and OTPs to keep false alarms low.
- Exported the model to a 500 KB format and ported the scorer to Kotlin for an offline Android app that flags incoming SMS and explains the reasons in English and Hindi.

**College Fee Management & Receipt System** — Team Project GitHub

*SAP ABAP, S/4HANA 1809, ABAP Dictionary, SNRO, ALV, CDS Views, OData, Smart Forms*

- Co-developed, in a team, an SAP ABAP system for college fee collection, payment verification and PDF receipts, with role-based access, ALV reports and OData integration.
- Designed the data model for 8 ABAP Dictionary tables and configured SNRO number ranges for unique payment IDs and receipt numbers.

## CERTIFICATIONS & ACTIVITIES

- Back-End Developer – ABAP Cloud — SAP ABAP training certificate (2026)
- Python — HackerRank Certificate · Python Course Completion Certificate
- Solved 50 problems on LeetCode and participated in CodeChef competitive programming contests.